

SGT HYDROEDGE

GREENLABS | SUSTAINABILITY INFRASTRUCTURE FOR INDUSTRY

CREDIBILITY & EVIDENCE DOSSIER Industrial HHO Generator — Kiln Applications



For Investors & Industrial Customers

2–9%

Fuel Efficiency Gain

**Up to
44%**

CO₂ Reduction

**Up to
80%**

NO_x Reduction

**Up to
70%**

PM Reduction

8–10%

Alt Fuel Use ↑



SECTION 1: THE INDUSTRIAL KILN PROBLEM

Industrial kilns are among the most energy-intensive and emission-heavy equipment in manufacturing. Whether firing ceramics, bricks, tiles, cement clinker, lime, or refractories, kilns consume enormous quantities of fuel — natural gas, coal, diesel, biomass, pet coke — yet deliver shockingly poor combustion efficiency.

KILN TYPE	TYPICAL FUEL EFFICIENCY	PRIMARY FUEL TYPES
Ceramic Roller Kilns	5–20%	Natural Gas, LPG
Brick / Batch Kilns	15–30%	Coal, Firewood, Gas
Tunnel Kilns	30–50%	Gas, Coal, Diesel
Rotary Cement Kilns	50–65%	Coal, Pet Coke, Alt Fuels
Lime / Calcination Kilns	40–60%	Gas, Coal, Oil

Every percentage point of efficiency gained translates to significant fuel savings and emission reductions at operating scale. The majority of input energy is currently lost through exhaust gases, radiation, and — critically — incomplete combustion. This is the precise problem that HHO oxyhydrogen injection solves.

SECTION 2: THE SGT HYDROEDGE SOLUTION

SGT HydroEdge is the first Indian company to manufacture high-capacity HHO (oxyhydrogen) generators engineered specifically for industrial kiln applications. Our approach is straightforward: generate hydrogen-oxygen gas on-demand through water electrolysis, and introduce it into the kiln's combustion zone as a combustion catalyst.

How It Works

HHO is a mixture of hydrogen and oxygen gas produced by electrolyzing water. When injected into a kiln's combustion zone in small, precise quantities:

- Hydrogen's flame propagation speed (~8–10x faster than natural gas) initiates more complete, faster combustion of the primary fuel
- Oxygen co-injected with hydrogen raises local oxygen availability at the combustion front, enabling more complete fuel burn
- The combined effect: higher flame temperatures, more complete combustion, lower fuel consumption, and dramatically fewer emissions
- No hydrogen storage is required — gas is produced on-demand and consumed immediately, eliminating explosion risk

SGT's Patent-Applied Technology Advantage

High HHO Yield	Patent-applied cell design delivers significantly higher hydrogen-oxygen gas output per unit of electrical input compared to conventional electrolyzers
Low Power Consumption	Optimised electrode geometry and electrolyte management reduce electricity consumption, ensuring net energy benefit is maximised
Long Operational Life	Engineered for continuous industrial duty. Durable materials and design choices ensure years of reliable operation without performance degradation
Safety by Design	On-demand generation means zero hydrogen storage on site. Built-in pressure relief, flashback arrestors, and fail-safe controls eliminate explosion risk
Retrofit — Zero Disruption	Installs on any existing kiln without modification to kiln structure, burners, or fuel lines. Minimal production downtime during installation
GreenVision Digital Twin	IoT-enabled monitoring integrates with SGT's AI-powered Digital Twin platform for real-time efficiency tracking, ESG reporting & carbon credit documentation
Made in India	Fully designed, developed, and manufactured by SGT HydroEdge in Pune. The only non-Chinese company manufacturing such industrial-scale HHO generators globally

SECTION 3: WHAT GLOBAL RESEARCH PROVES

Researchers across the world — at leading universities, industry bodies, and government-funded programmes — have extensively studied hydrogen-enriched combustion in kilns. The findings are consistent across kiln types and fuel types: efficiency improves, emissions fall, product quality is maintained.

3.1 Ceramic Tile Kilns (Continuous Roller Kilns)

- Hydrogen-enriched combustion experiments on an industrial ceramic kiln with 102 burners (Tsinghua University, China) demonstrated optimised energy efficiency of 63.13% for tile production, with reduced natural gas consumption at optimum hydrogen flow rates
- Ceramic roller kilns typically operate at very low energy efficiency (5–20%). Energy savings of up to 17% have been documented through improved combustion enabled by hydrogen
- An EU-funded project (LIFE ZEF-tile, Spain/Italy) demonstrated oxygen-enriched firing in ceramic tile kilns reduced CO₂ emissions by 90%, NO_x by 80%, and particulate matter by 70%, while consuming approximately 10% less fuel
- European kiln manufacturers (SACMI, Italy) report that advanced combustion for tile kilns delivers 8–10% natural gas savings — HHO-assisted combustion achieves comparable or better results as a retrofit to existing kilns without burner replacement
- Multiple hydrogen combustion trials confirmed that ceramic tile quality remains consistent with conventional firing when combustion parameters are properly managed (ITC-AICE, Spain; Lucideon, UK)

3.2 Batch Kilns (Bricks, Pottery & Refractories)

- Batch kilns using firewood, coal, or gas experience significant incomplete combustion, especially during low-temperature drying stages where CO emissions are highest. HHO injection improves combustion completeness across all temperature stages
- The British Ceramic Confederation's feasibility programme demonstrated successful hydrogen firing across heavy clay (bricks), whitewares, and refractories — products met all technical specifications. Both batch and continuous kiln configurations were tested
- Research confirms that hydrogen's faster flame propagation and lower ignition energy enable more uniform temperature distribution inside batch kilns, reducing hot spots and cold zones that cause product defects and energy waste
- In brick kilns where firewood is the primary fuel, HHO injection reduces the total firewood requirement and dramatically cuts CO emissions during the critical low-temperature drying phase

3.3 Cement Kilns (Rotary Kilns) — World-Leading Real Deployments

★ **CEMEX — Global Leader (Spain, Europe, Mexico, Dominican Republic, Honduras)**

CEMEX, one of the world's largest cement producers (NYSE: CX), is the most significant real-world validation of hydrogen injection in kilns at global scale:

- Trial initiated at Alicante Plant, Spain — July 2019. Confirmed immediate potential as a CO₂ reduction lever
- By 2020, hydrogen injection technology installed across ALL CEMEX cement plants in Europe
- By 2021, rolling out to substantially all global operations — Mexico, USA, South & Central America, Caribbean, Africa, Asia (US\$40M programme)
- Technology increases alternative fuel substitution rates by 8–10 percentage points and has cut thousands of tonnes of CO₂ annually
- **Methodology: On-site electrolysis splits water; hydrogen AND oxygen injected directly into kiln main burner — exactly as SGT HydroEdge's approach**

Source: CEMEX official press releases (cemex.com), Global Cement Magazine, GCCA

★ Cementos Argos — Honduras (2022)

- Hydrogen and oxygen from electrolysis injected directly into the kiln main burner at the Piedras Azules plant
- Result: 3% rise in clinker throughput and 2% drop in specific fuel consumption
- **Internal Rate of Return on pilot investment: 204% — commercial viability conclusively demonstrated**

Source: Global Cement Magazine, September 2022; PACA Web

★ Hanson / Heidelberg Materials — Ribblesdale, UK (2021)

- UK Government-funded Fuel Switching Project. Full-scale rotary kiln with four-stage pre-heater at Ribblesdale plant
- Achieved net-zero CO₂ fuel emissions from the main burner during trial periods using hydrogen + biomass blend
- Clinker and cement quality confirmed unchanged — verified by VDZ (German Cement Works Association) via XRD, XRF and microscopy analysis
- Temperature profile and coating formation remained stable throughout — no adverse kiln process effects

Source: VDZ / Global Cement Magazine, 'H₂ in Cement: Lessons Learned', 2022; UK BEIS Fuel Switching Project

★ Tarmac (CRH) — Tunstead Lime Plant, UK (2022)

- Successful trial using hydrogen as an alternative to natural gas at Tunstead lime plant (July 2022)
- Demonstrated hydrogen viability in lime kiln applications — extending the technology proof beyond cement to lime and calcination

Source: Global Cement Magazine, September 2022

★ Limak Cement — Turkey (Calcliner Application)

- Pioneered hydrogen injection into the calciner using a 50% hydrogen-rich mix — the lower-temperature pre-processing stage of cement production that accounts for a large share of fuel demand in modern preheater lines
- **Limak aims to run all its plants on hydrogen-enhanced fuel by 2030, with projected savings of 700,000 tonnes CO₂ per year**

Source: *Hydrogenera.eu, H2 Bulletin*

3.4 Industrial Burner Research — Direct HHO Validation

- A 2025 peer-reviewed study (ScienceDirect, Fuel journal) directly tested a scalable multi-cell dry electrolyser coupled to an industrial natural-gas burner with 40,000 kcal/h thermal capacity — a direct parallel to kiln burner applications
- Results: 25% reduction in CO emissions, 7% reduction in unburned hydrocarbons, and 8–9°C increase in flame temperature with HHO enrichment. Combustion efficiency rose to 80.6%
- Optimum operation: 10–15A (400W) produced 130 L/h HHO with 2 Wh/L energy intensity — confirming the energy economics of on-site HHO production

Source: *ScienceDirect / Fuel Journal, 2025* — 'Experimental investigation of hydroxy gas as supplementary fuel for industrial burners'

3.5 Thermodynamic Research — Cement Kiln Oxygen Enrichment

- Research published in MDPI (Processes, 2024) on oxygen-enriched combustion in cement kilns shows: exergy efficiency of the complete kiln system rises by 17.56% vs conventional processes; coal usage cut by 30%; exhaust gas flow reduced by 54.62%
- HHO injection provides BOTH hydrogen (combustion catalyst) AND oxygen (enrichment) simultaneously — delivering dual thermodynamic benefit in a single system
- Thermodynamic analysis confirms hydrogen-based fuel in cement production can reduce CO₂ from fuel combustion by up to 44% vs conventional coal firing

Source: *MDPI Processes, 2024* — 'Coupled Oxygen-Enriched Combustion in Cement Industry CO₂ Capture System'

SECTION 4: MARKET CONTEXT & SGT'S UNIQUE POSITION

The Technology Is Proven — The Market Needs Indian Supply

Industrial-scale HHO generators for kiln and combustion applications have been manufactured and deployed commercially — primarily by Chinese companies — for several years. The technology is not experimental. The business case is established. The question for Indian industry is: who do you buy from?

FACTOR	CHINESE SUPPLIERS	SGT HYDROEDGE
Manufacturing Base	China	Pune, India — Made in India
After-Sales Support	Limited, high response time	On-ground Indian team, fast response
Customisation	Standard models	Customised for your kiln type & fuel
Supply Chain Risk	Import dependencies, geopolitical risk	Domestic supply, no import risk
Regulatory Compliance	CE-marked, international	BIS compliant, India-specific safety
Financing & Leasing	Not available	India-specific models possible
Technology IP	Chinese-owned	Indian patent-applied, SGT-owned
Digital Integration	Basic monitoring	GreenVision AI Digital Twin, IoT-native

SGT HydroEdge is today the only non-Chinese company manufacturing industrial-scale HHO generators with proven technology. For Indian kiln operators — in ceramics, bricks, cement, lime, and refractories — this represents a strategically superior option: proven technology, domestic supply, full support, Indian IP.

SECTION 5: SGT HHO GENERATOR — KILN SPECIFICATION

SPECIFICATION	SGT HYDROEDGE HHO GENERATOR FOR KILNS
HHO Output	Up to 3000 LPH (scalable via modular design for higher requirements)
Technology	Liquid Cell — high efficiency, low maintenance, long operational



	life
Power Supply	10 kVA IGBT-based high-efficiency power supply
Operational Capability	24×7 continuous; up to 7 days non-stop operation
Safety Features	Multiple electrical & process interlocks, flashback arrestors, pressure relief, fail-safe shutdown
Hydrogen Storage	Zero — on-demand generation only. No explosive gas inventory on site
Installation	Retrofit — no kiln or burner modification required
Connectivity	IoT-enabled. Integrates with GreenVision Digital Twin for real-time monitoring
Kiln Types Supported	Batch kilns, continuous roller kilns, tunnel kilns, rotary cement kilns, lime kilns
Fuel Types Supported	Natural gas, CNG, coal, pet coke, diesel, biomass, firewood, alternative/waste-derived fuels
Industries Served	Ceramics, tiles, bricks, cement, lime, refractories, glass, calcination
Origin	Designed, developed & manufactured by SGT HydroEdge in Pune, India
IP Status	Patent applied. Proprietary cell design and system architecture

SECTION 6: INDICATIVE ROI MODEL — KILN OPERATOR

The following is an indicative financial model for a mid-size industrial kiln operator to illustrate the economics of SGT HydroEdge deployment. Actual results will vary based on kiln type, fuel type, operating hours, and fuel prices.

PARAMETER	VALUE (ILLUSTRATIVE)
Annual fuel spend (kiln)	₹1,00,00,000 (₹1 Crore)
Fuel savings @ 5% efficiency improvement	₹5,00,000 per annum
Emission reduction (CO₂)	Up to 44% reduction on fuel-combustion CO ₂
HHO system power consumption	~10 kVA (included in operating cost)
Typical payback period	12–24 months (fuel savings alone)
Additional benefit: Carbon credits	Potential additional revenue stream via verified emission reductions
Additional benefit: ESG compliance	Automated ESG/CBAM reporting via GreenVision platform

SECTION 7: GLOBAL RESEARCH & SOURCE REFERENCES

The evidence base for hydrogen-enriched combustion in kilns is substantial, peer-reviewed, and global. Below are the key research sources and institutions whose work validates this technology:

CATEGORY	INSTITUTION / COMPANY	COUNTRY / PUBLICATION
Ceramic Kilns	Tsinghua University — industrial ceramic kiln hydrogen combustion study	China / Peer-reviewed journal
Ceramic Kilns	Institute of Ceramic Technology (ITC-AICE)	Spain / Industry research
Ceramic Kilns	SACMI — advanced burner technology for tile kilns	Italy / Industry publication
Ceramic Kilns	EU LIFE ZEF-tile Project	EU / European Commission LIFE Programme
Ceramic Kilns	Lucideon — hydrogen firing feasibility for ceramics	UK / Industry research
Batch / Brick Kilns	British Ceramic Confederation — hydrogen firing feasibility	UK / Industry programme
Cement Kilns	CEMEX — hydrogen injection all European plants	Global (NYSE: CX) / Official press releases
Cement Kilns	Hanson / Heidelberg Materials — Ribblesdale UK trial	UK / VDZ / Global Cement Magazine

Cement Kilns	Cementos Argos — Piedras Azules, Honduras	Honduras / Global Cement Magazine
Cement Kilns	VDZ — German Cement Works Association	Germany / Industry report
Cement Kilns	MDPI Processes — 'Oxygen-Enriched Combustion in Cement Kilns'	Global / MDPI (Peer-reviewed, 2024)
Lime Kilns	Tarmac (CRH) — Tunstead lime plant hydrogen trial	UK / Global Cement Magazine
Lime / Calcination	Limak Cement — calciner hydrogen injection	Turkey / Industry press
Industrial Burners	ScienceDirect / Fuel Journal — HHO in 40,000 kcal/h industrial burner	Global / Elsevier Fuel Journal (2025)
Thermodynamics	SpringerLink — 'High Efficiency Zero Carbon Emission HHO Generator'	India/Global / Springer (2024)

Ready to Improve Your Kiln Efficiency & Cut Emissions?

Ask us for a no-obligation kiln assessment and ROI calculation.

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